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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 07/09/2007 |
| **PI Name:** | Triplett, Jason Ph.D. | | |
| **Project Title:** | Investigating the role of alpha-actinin in osteoblast mechanotransduction | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Developmental biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 46202 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | GSRP |
| **Start Date:** | 07/01/2004 | **End Date:** | 06/30/2007 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA HQ |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Bone loss associated with exposure to microgravity occurs due to a lack of mechanical loading. The cellular mechanisms by which bone cells sense and respond to mechanical stimuli remain unclear. Our laboratory has shown that focal adhesions play an important role in mechanotransduction in osteoblasts. I am proposing to study the role of alpha-actinin in osteoblast mechanotransduction. Based on alpha-actinin's ability to interact with multiple focal adhesion proteins, it could potentially play an important role in focal adhesion organization and mechanotransduction. I will use a dominant-negative alpha-actinin to assess alpha-actinin's role in focal adhesion organization and mechanotransduction. | | |
| **Rationale for HRP Directed Research:** | | | |

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| **Research Impact/Earth Benefits:** | Understanding how bone cells sense and respond to mechanical stimuli can help to identify molecular targets to offset the loss associated with long-term space flight. Any manned mission to Mars, for instance, will need to address the issue of bone loss associated with a micro-gravity environment. This may require a combination of molecular medicine and the physical generation of "artificial gravity." |
| **Task Progress:** | Within the past year, our lab has made significant progress towards the completion of the aims of our research proposal. First, we have published a paper in which we explored the effects of expression of a truncated version of a-actinin on focal adhesion organization and cell survival (Triplett and Pavalko. Am J Physiol – Cell Physiol, 2006 Nov;291(5):C909-21). This construct, termed ROD-GFP, effectively disrupts focal adhesion organization, as evidenced by a reduction in focal adhesion size. ROD-GFP expression also results in inhibited focal adhesion signaling capabilities, evidenced by decreased tyrosine phosphorylation and focal adhesion kinase (FAK) phosphorylation. Further more, we demonstrated that expression of ROD-GFP renders cells more susceptible to apoptosis induced by tumor necrosis factor alpha (TNFa). We showed that this sensitization was due to decreased expression of anti-apoptotic proteins, such as Bcl-2 and Bcl-XL, and decreased survival signaling through the TNFa receptor pathway. These results are of particular interest, because regulation of apoptosis at sites of bone formation have been shown to be important in the ability of bone to respond to mechanical stimuli properly. Additionally, a careful examination of the role of a-actinin in fluid flow induced signaling was undertaken. We previously found that cells expressing ROD-GFP showed reduced ERK and p38 MAP kinase activation in response to fluid shear stress. Despite this, we observed no difference in the ability of ROD-GFP expressing cells to upregulate cyclo-oxygenase-2, a key enzyme in bone formation. These results were confirmed by western blot and by quantitative RT-PCR. Furthermore, we observed no difference in flow-induced prostaglandin-E2 release in ROD-GFP-expressing cells. Interestingly, we did observe an inhibition of flow induced osterix expression in cells expressing ROD-GFP. Osterix is an important transcription factor involved in the maturation and differentiation of osteoblasts into osteocytes. These data suggest that a-actinin may play an important role in early responses to fluid shear, but only a selective role in events further downstream.<br>Finally, through collaborative efforts with fellow members of the Pavalko lab, we were able to publish two papers this year, further extending our understanding of the mechanisms of mechanically-induced bone formation. In the first paper (J Cell Biochem, 2007 Feb 15;100(3):794-807), we examined the differential effects of unidirectional and oscillatory fluid shear stress in osteoblasts and osteocytes. Using immortalized cell lines, we found that both osteoblasts and osteocytes formed stress fibers in response to unidirectional fluid shear stress. However, osteocytes did not form stress fibers in response to oscillatory fluid shear, and, in fact, adopted a more dendritic morphology. Despite these differences, we found that both cell types were able to upregulate both COX-2 and osteopontin. We concluded that subtypes of bone cells can detect differences in flow profile, and are able to respond differently as well. In the second paper (Mol Cell Biomech, in press), we examined the effect of fluid shear stress on growth factor signaling in cultured osteoblasts. We found that cells subjected to fluid shear stress respond more robustly to insulin-like growth factor treatment in the form of increased ERK and Akt kinase phosphorylation, as well as increased resistance to apoptotic stimuli. These results indicate a cross-talk between mechanical and growth factor signaling pathways, which may explain the pro-survival effects of fluid shear stress. In conclusion, our studies over the past year have yielded interesting and important results concerning the role of a-actinin and focal adhesions in mechanotransduction. As a result of the work produced under this NASA funded fellowship, we have published four papers in peer-reviewed journals, and I have successfully defended my thesis. We believe that our efforts have provided the field with novel tools to aid in researching the role of cell adhesion in osteoblast physiology and advanced our understanding of mechanotransduction, which could lead to the development of countermeasures to the loss of bone associated with spaceflight. Finally, I appreciate the support from NASA in the pursuit of this project. |
| **Bibliography Type:** | Description: (Last Updated: 01/08/2020) |
| **Articles in Peer-reviewed Journals** | Ponik SM, Triplett JW, O'Riley RL, Pavalko FM. "Osteoblasts and osteocytes respond differently to oscillatory and unidirectional fluid flow profiles." J Cell Biochem. 2007 Feb 15;100(3):794-807. PMID: 17031855 , Feb-2007 |
| **Articles in Peer-reviewed Journals** | Triplett JW, O'Riley RL, Tekulve K, Norvell SM, Pavalko FM. "Mechanical loading by fluid shear stress enhances IGF-1 receptor signaling in osteoblasts in a PKCzeta-dependent manner." Mol Cell Biomech. 2007 Mar;4(1):13-25. PubMed PMID: 17879768 [NOTE reported previously in July 2007 as "in press"] , Mar-2007 |
| **Articles in Peer-reviewed Journals** | Triplett JW, Pavalko FM. "Disruption of alpha-actinin-integrin interactions at focal adhesions renders osteoblasts susceptible to apoptosis." Am J Physiol Cell Physiol. 2006 Nov;291(5):C909-21. Epub 2006 Jun 28. PMID: 16807302 , Nov-2006 |